

Cancer Genomics & Bioinformatics

Day 2: Alignments & mapping

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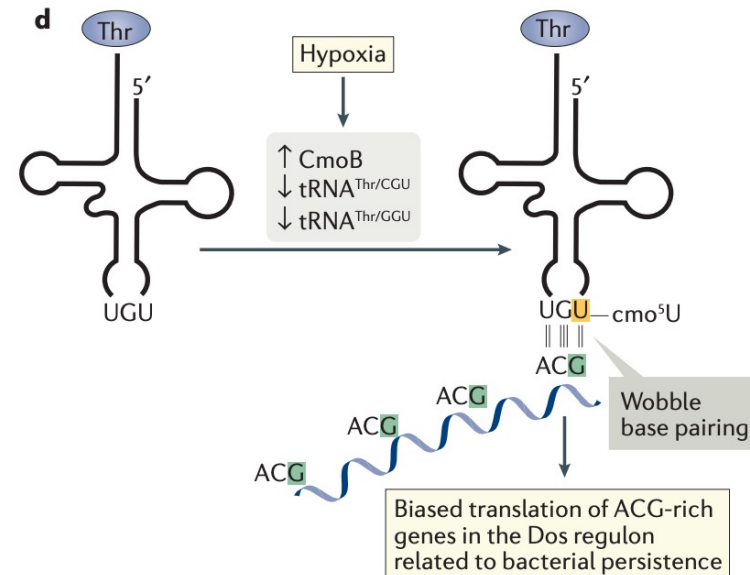
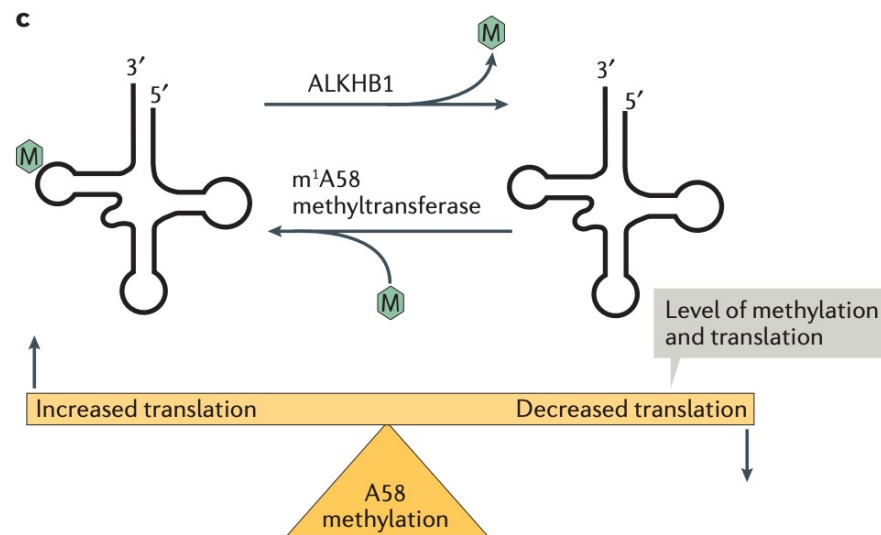
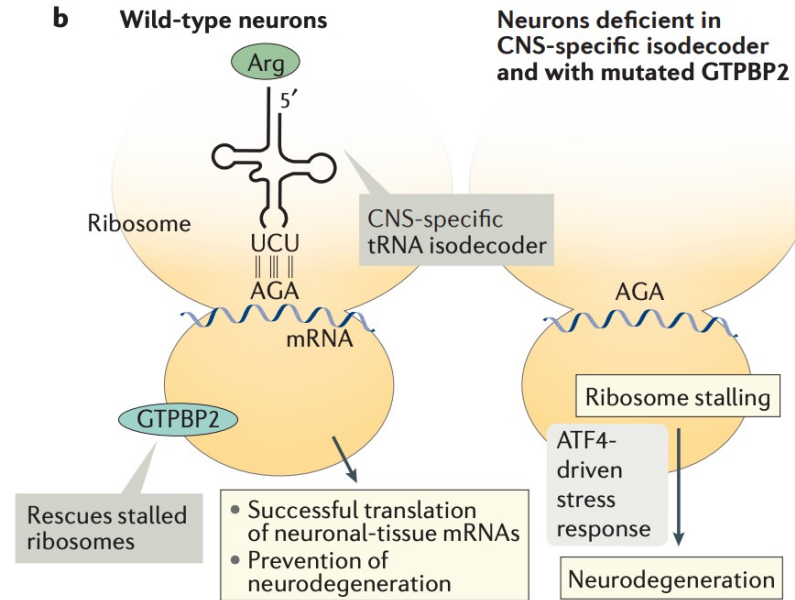
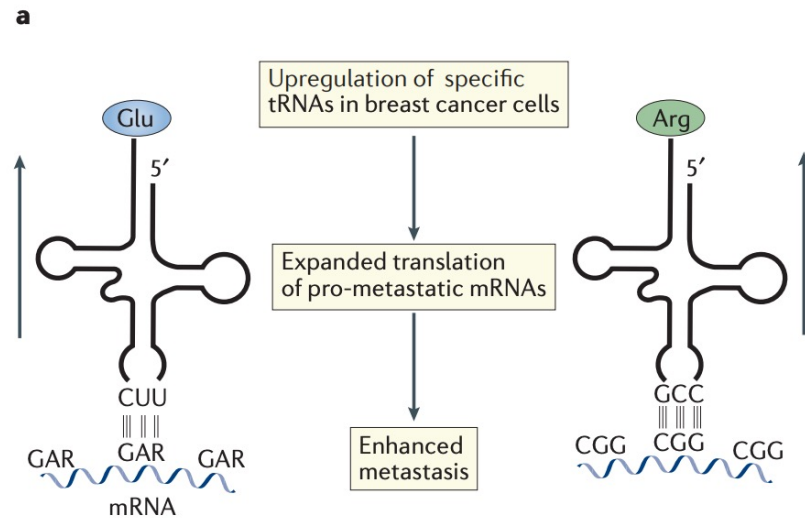
University of Miami Miller school of Medicine

<https://github.com/atelonis/BioinformaticsExercises>

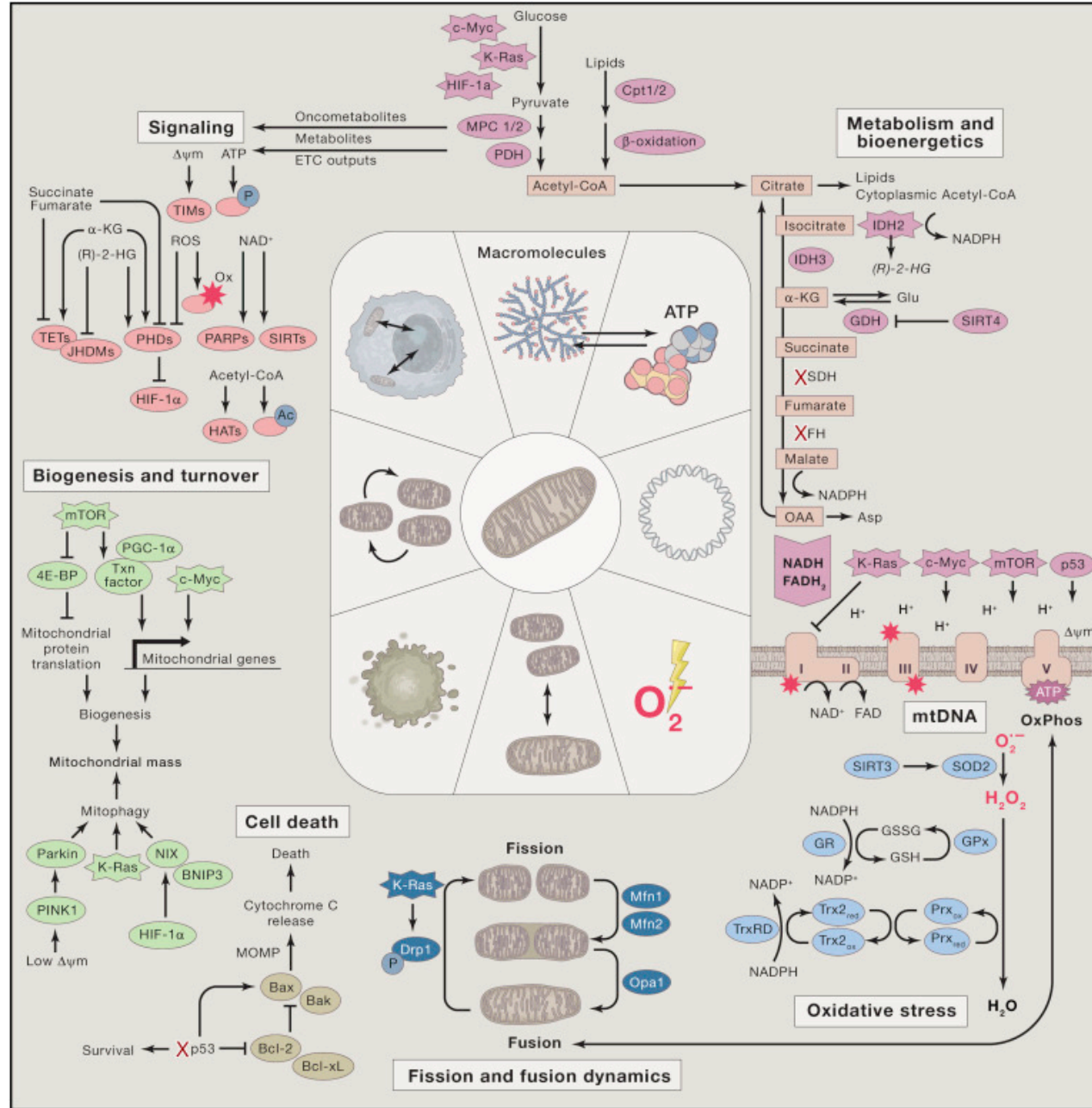
Learning objectives

- How/where are biological sequences stored?
- How can I align one biological sequence on another?
- How can I merge/intersect genomic coordinates (genome arithmetic)?
 - What is a “local” and what is a “global” alignment?
 - What applications use sequencing alignment?
- What are the first steps in analyzing RNA-sequencing data?
- How is mapping different from transcript quantification?

tRNAs are not just for delivery of amino acids

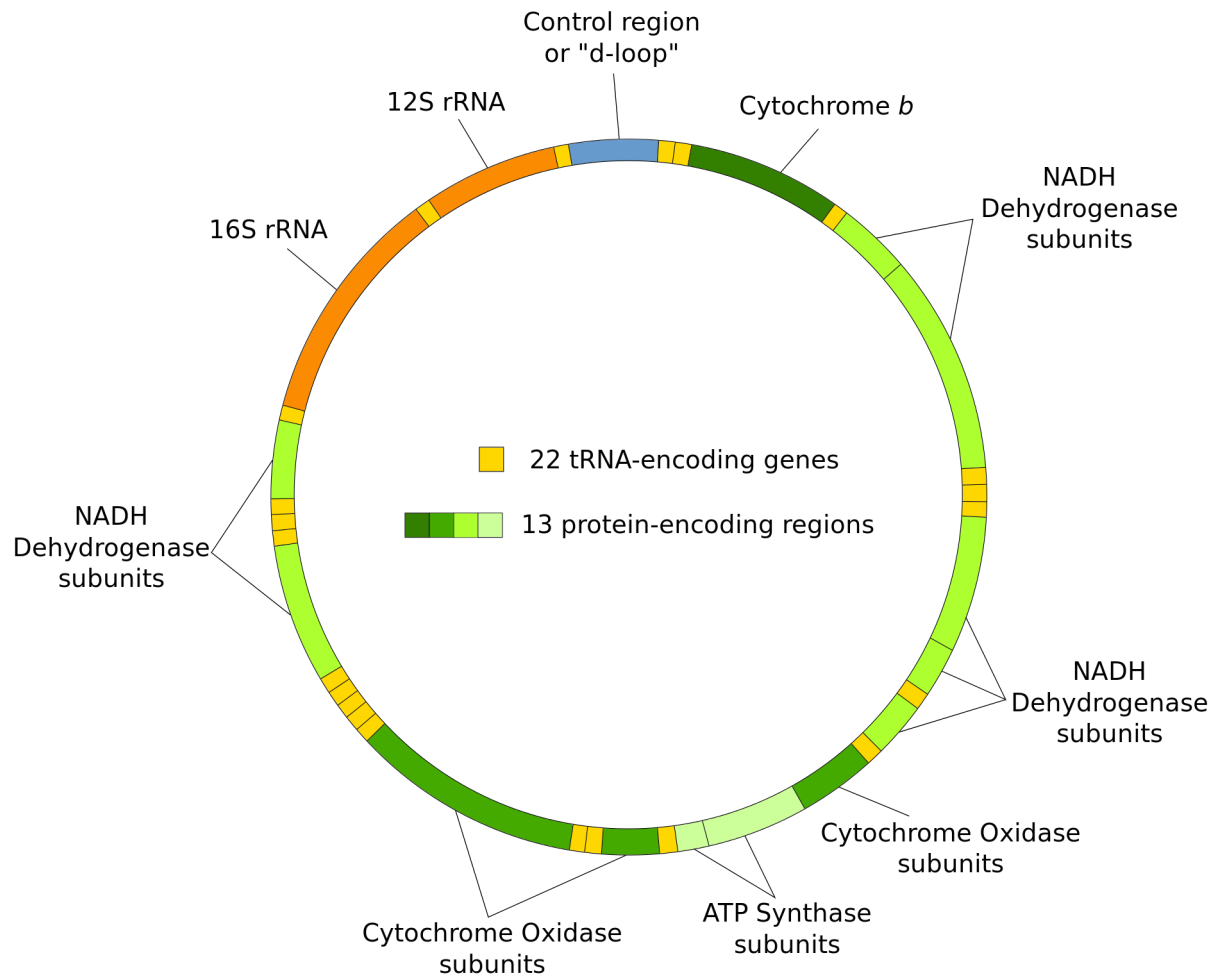


Mitochondria are important mediators of tumorigenesis

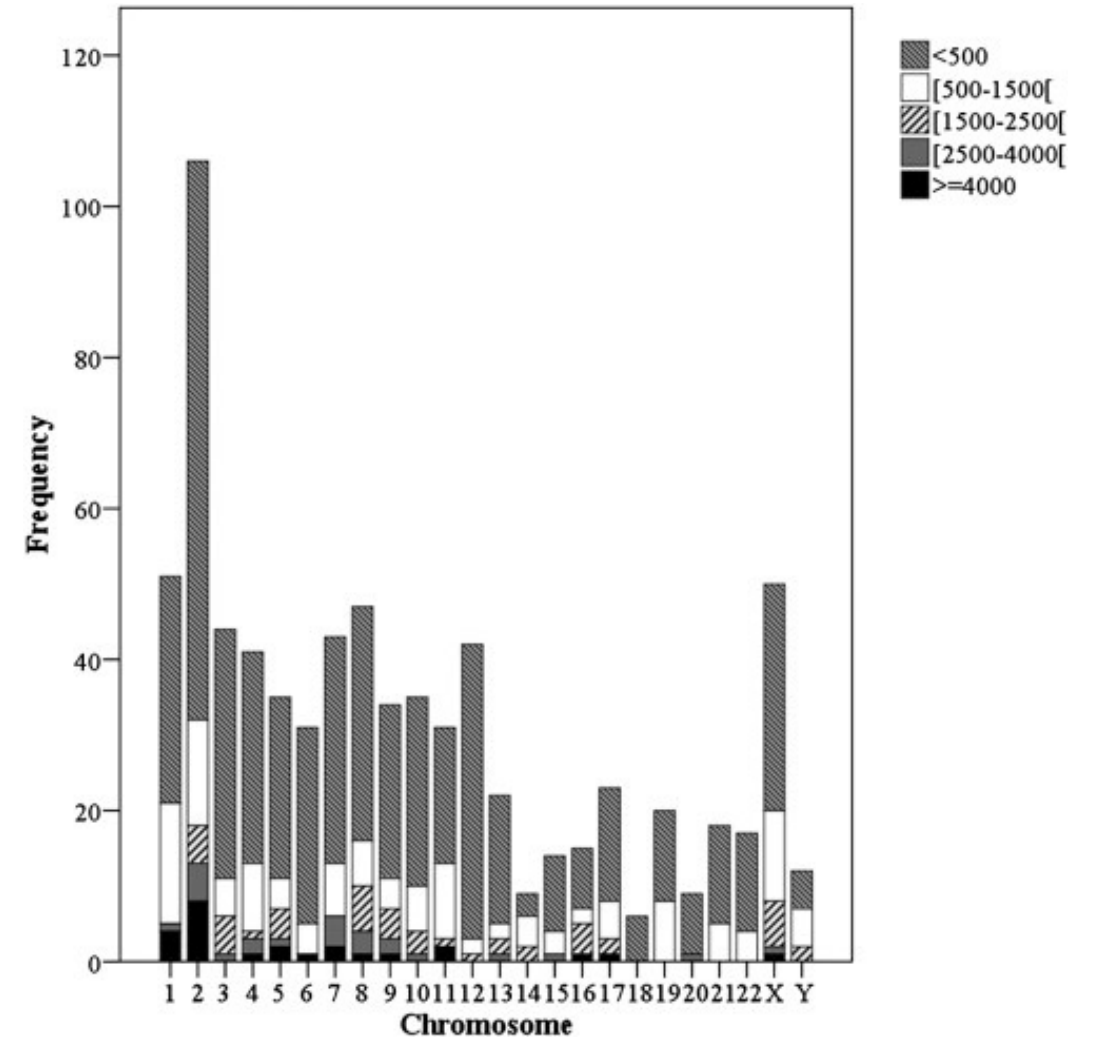


Vyas et al. (2016)
Cell 166(3):555-566

The mitochondrial genome

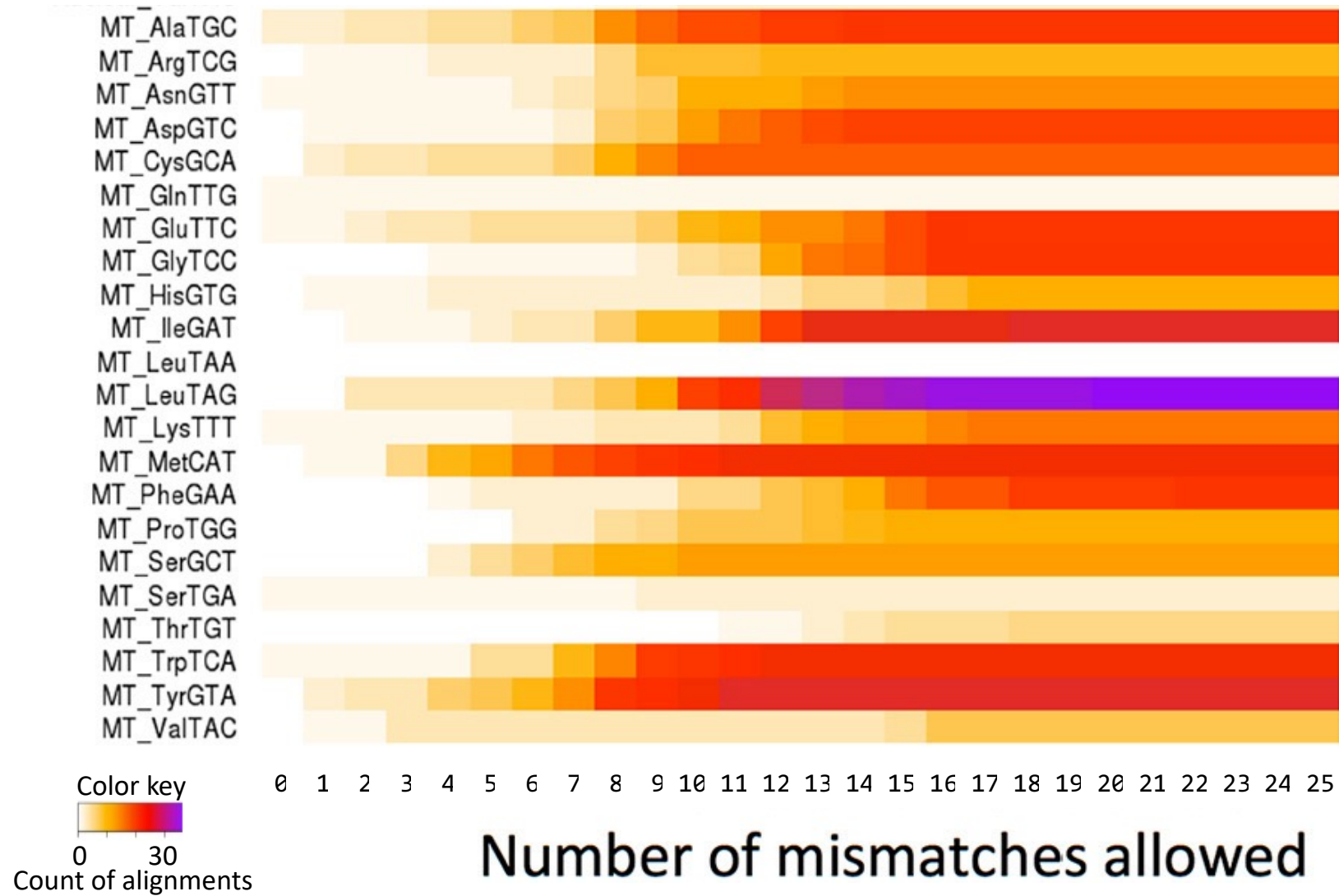


By derivative work: Shanel (talk) Mitochondrial DNA de.svg: translation by Knopfkind; layout by jhc - Mitochondrial DNA de.svg, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=4304443>



Frequency and size (bp) of insertions for each chromosome.
Ramos et al. (2011) *Mitochondrion* 11(6):946-53

LeuTAA tRNA does not have nuclear copies



For further reading:

Ramos et al. (2011) Nuclear insertions of mitochondrial origin. Database updating and usefulness in cancer studies. Mitochondrion 11(6):946-53

<https://pubmed.ncbi.nlm.nih.gov/21907832/>

Telonis et al. (2014) Nuclear and mitochondrial tRNA-lookalikes in the human genome. Front Genet 5:344. <https://pubmed.ncbi.nlm.nih.gov/25339973/>

Telonis et al. (2015) Mitochondrial tRNA-lookalikes in nuclear chromosomes: could they be functional? RNA Biol 12(4):375-80. <https://pubmed.ncbi.nlm.nih.gov/25849196/>

Tsuji et al. (2012) Mammalian NUMT insertion is non-random. Nucleic Acids Res 40(18):9073-9088. <https://pubmed.ncbi.nlm.nih.gov/22761406/>

<https://en.wikipedia.org/wiki/NUMT>

-- Our goal --

Look for mitochondrial tRNAs on nuclear chromosomes

We need:

- ☐ A file with the sequences of nuclear chromosomes
- ☐ A file with the sequences of the mitochondrial tRNAs
- ☐ A tool to do the alignments (blastn)
- ☐ The positions of known tRNAs on nuclear chromosomes

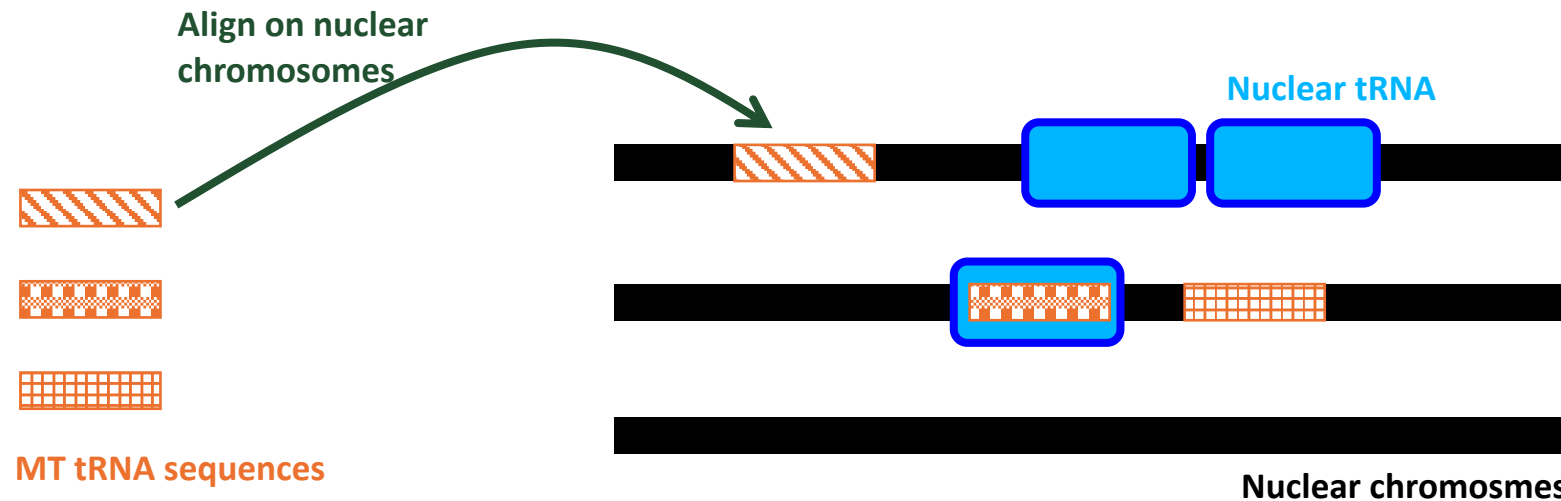
Sources:

UCSC Genome browser

ENSEMBL/BioMart

blastn

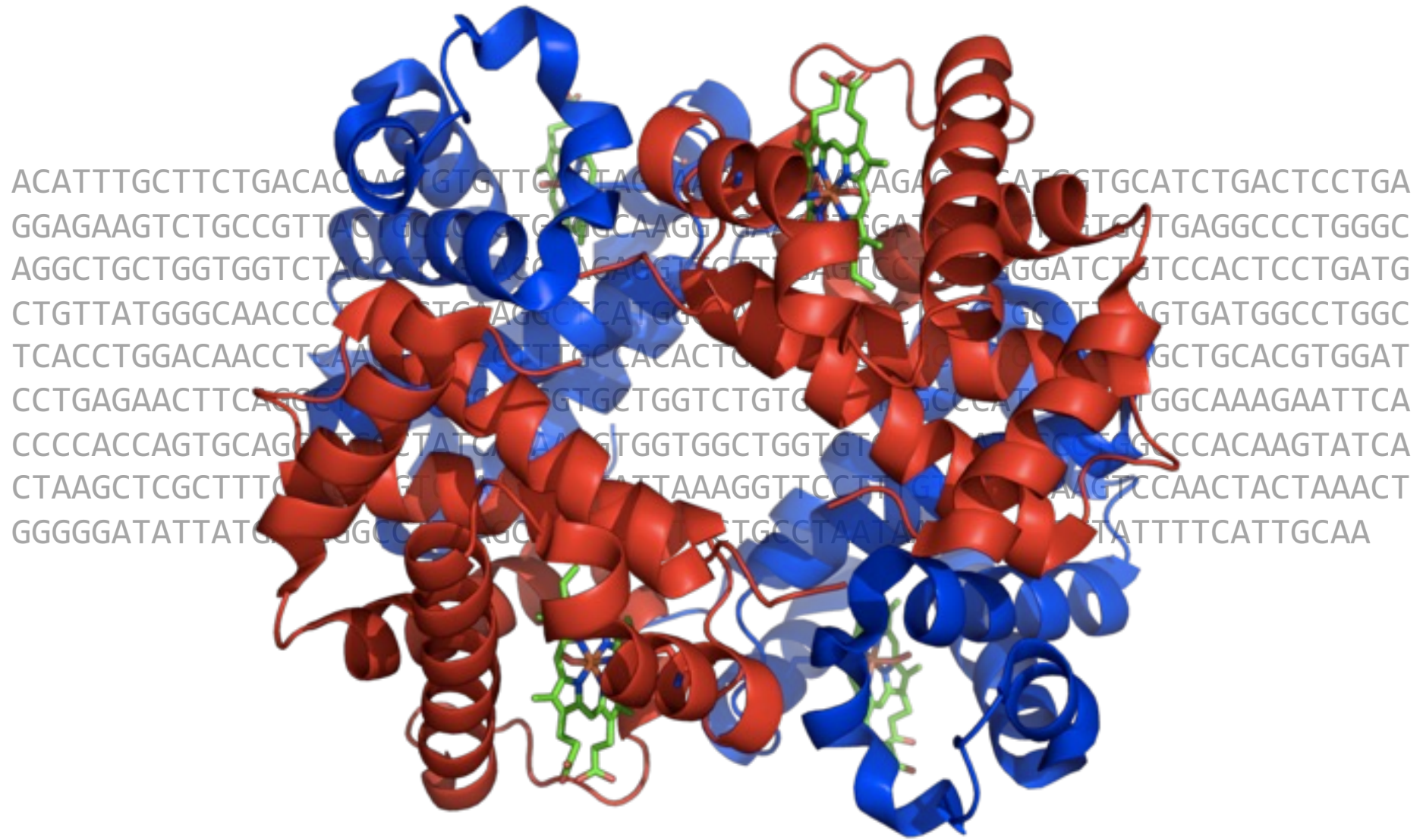
UCSC Genome browser



ACATTTGCTTCTGACACAACCTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGA
GGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGC
AGGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGC
TCACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGAT
CCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCA
CCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCA
CTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCTTTGTTCCCTAAGTCCAACCTACTAACT
GGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAA

+ 6.4 billion letters

DNA is a sequence of four letters



BLAST: the “google” of the genome

**U.S. National Library of Medicine**
National Center for Biotechnology Information

Log in

BLAST®

HomeRecent ResultsSaved StrategiesHelp

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

A new feature was added to Primer-BLAST.

We now offer the ability for user to run primer-blast from NCBI assembly page..

Tue, 23 Feb 2021 12:00:00 EST

[More BLAST news...](#)

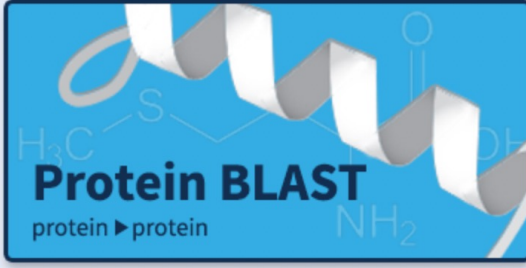
Web BLAST



Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide



Protein BLAST
protein ► protein

BLAST Genomes

HumanMouseRatMicrobes

Search

<https://blast.ncbi.nlm.nih.gov/>

If you didn't trust me that sequence was HBB

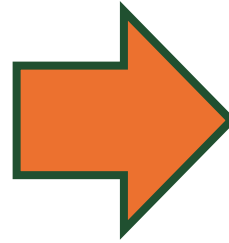
Descriptions									
Graphic Summary									
Alignments									
Taxonomy									
Sequences producing significant alignments									
Download New Select columns Show 100 ?									
☒ select all 100 sequences selected									
GenBank Graphics Distance tree of results New MSA Viewer									
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Homo sapiens hemoglobin subunit beta (HBB), mRNA	Homo sapiens	1160	1160	100%	0.0	100.00%	628	NM_000518.5
☒	PREDICTED: Pan troglodytes hemoglobin subunit beta (HBB), mRNA	Pan troglodytes	1155	1155	100%	0.0	99.84%	748	XM_508242.4
☒	PREDICTED: Pan paniscus hemoglobin subunit beta (LOC100976465), mRNA	Pan paniscus	1149	1149	100%	0.0	99.68%	646	XM_003819029.3
☒	Homo sapiens hemoglobin, beta, mRNA (cDNA clone MGC:14540 IMAGE:4292125), complete cds	Homo sapiens	1142	1142	99%	0.0	99.52%	658	BC007075.1
☒	Human messenger RNA for beta-globin	Homo sapiens	1140	1140	99%	0.0	99.52%	626	V00497.1
☒	PREDICTED: Gorilla gorilla gorilla hemoglobin subunit beta (LOC101126932), mRNA	Gorilla gorilla go...	1133	1133	100%	0.0	99.20%	753	XM_019036164.1
☒	Homo sapiens hemoglobin beta mRNA, complete cds	Homo sapiens	1133	1133	99%	0.0	99.36%	647	AY509193.1
☒	PREDICTED: Pongo abelii hemoglobin subunit beta (HBB), mRNA	Pongo abelii	1098	1098	99%	0.0	98.25%	627	XM_002822127.4
☒	PREDICTED: Nomascus leucogenys hemoglobin subunit beta (HBB), mRNA	Nomascus leuc...	1088	1088	100%	0.0	97.93%	753	XM_004090649.3
☒	PREDICTED: Hylobates moloch hemoglobin subunit beta (HBB), mRNA	Hylobates moloch	1077	1077	100%	0.0	97.61%	753	XM_032166808.1
☒	Homo sapiens hemoglobin beta chain variant (HBB) mRNA, HBB-Dothan allele, complete cds	Homo sapiens	1064	1064	93%	0.0	99.32%	587	EU694432.1
☒	Homo sapiens hemoglobin beta subunit variant (HBB) mRNA, complete cds	Homo sapiens	1064	1064	92%	0.0	99.83%	579	AF181989.1
☒	Homo sapiens beta globin chain variant (HBB) mRNA, complete cds	Homo sapiens	1057	1057	92%	0.0	99.48%	581	AF349114.1
☒	PREDICTED: Trachypithecus francoisi hemoglobin subunit beta (LOC117081249), mRNA	Trachypithecus f...	1038	1038	100%	0.0	96.51%	749	XM_033207068.1
☒	PREDICTED: Colobus angolensis palliatus hemoglobin, beta (HBB), mRNA	Colobus angole...	1038	1038	100%	0.0	96.50%	753	XM_011963601.1
☒	PREDICTED: Rhinopithecus roxellana hemoglobin subunit beta (LOC104662339), mRNA	Rhinopithecus r...	1033	1033	100%	0.0	96.35%	749	XM_010363344.2
☒	PREDICTED: Rhinopithecus bieti hemoglobin subunit beta (LOC108521664), mRNA	Rhinopithecus b...	1033	1033	100%	0.0	96.35%	749	XM_017861621.1
☒	Homo sapiens hemoglobin beta chain (HBB) mRNA, complete cds	Homo sapiens	1027	1027	89%	0.0	99.64%	562	AF117710.1
☒	Human sickle beta-hemoglobin mRNA	Homo sapiens	1024	1024	91%	0.0	98.44%	576	M25113.1
☒	PREDICTED: Mandrillus leucophaeus hemoglobin subunit beta (LOC105535916), transcript variant X2, mRNA	Mandrillus leuco...	1022	1022	100%	0.0	96.02%	753	XM_0119751...
☒	PREDICTED: Ptilinopus tenipectus hemoglobin subunit beta (HBB), mRNA	Ptilinopus ten...	1016	1016	100%	0.0	95.87%	748	XM_0232096...



What you need to run blast: a query and a search database

Query file

```
>MyQuery
ACATTTGCTTCTGACACAACACTGTGTTCACTAG
CAACCTCAAACAGACACCATGGTGCATCTGAC
TCCTGAGGAGAAGTCTGCCGTTACTGCCCTGT
GGGGCAAGGTGAACGTGGATGAAGTTGGTGGT
GAGGCCCTGGGCAGGCTGCTGGTGGTCTACCC
TTGGACCCAGAGGTTCTTTGAGTCCTTTGGGG
ATCTGTCCACTCCTGATGCTGTTATGGGCAAC
CCTAAGGTGAAGGCTCATGGCAAGAAAGTGCT
CGGTGCCTTTAGTGATGGCCTGGCTCACCTGG
ACAACCTCAAGGGCACCTTTGCCACACTGAGT
GAGCTGCACTGTGACAAGCTGCACGTGGATCC
TGAGAACTTCAGGCTCCTGGGCAACGTGCTGG
TCTGTGTGCTGGCCCATCACTTTGGCAAAGAA
TTCACCCCACCAGTGCAGGCTGCCTATCAGAA
AGTGGTGGCTGGTGTGGCTAATGCCCTGGCCC
ACAAGTATCACTAAGCTCGCTTTCTTGCTGTC
CAATTTCTATTAAAGGTTCCCTTTGTTCCCTAA
GTCCAACACTAACTGGGGGATATTATGAAG
GGCCTTGAGCATCTGGATTCTGCCTAATAAAA
AACATTTATTTTCATTGCAA
```



Database

```
>chr1
ACAACTGTGTTCACTAGCAAACATTTGCTTCTGAC.....
>chr2
ACATTTGCTTCTGACACAACACTGTGTTCACTAGCAA.....
.....
>chrX
TCTGACACAACACTGTGTTCACTAGCAAACATTTGCT.....
```

Fasta files (.fa) are used to store sequences in a simple format: Lines starting with ">" contain the name, description or any other information about the sequence and are followed by the sequence (in one or multiple lines) until the next ">" (if any).

Where can I find the human genome?

→ ENSEMBL

→ UCSC Genome Browser

<http://genome.ucsc.edu/> > Downloads > Genome Data > Human > Sequence data
by chromosome [for GRCh38]

Download the following:

chr1.fa.gz

chr6.fa.gz

chrM.fa.gz

We will not search the whole genome but instead only chromosomes 1 and 6 for speed and simplicity. We also include the mitochondrial genome as a “positive control” – our methodology should find the MT ones, otherwise there is something wrong.

Where can I find the MT tRNA sequences?

The screenshot shows the Ensembl BioMart interface. At the top, the Ensembl logo is on the left, and navigation links (BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, Blog) and a search bar are on the right. Below the navigation bar, there are tabs for 'New', 'Count', and 'Results'. The 'Results' tab is active, showing a list of results. On the left side of the results table, there are sections for 'Dataset' (Human genes (GRCh38.p13)), 'Filters' (Gene type: Mt_tRNA), and 'Attributes' (Gene stable ID, Gene name, Unspliced (Transcript)). The 'Export' section at the top right of the results table allows selecting the format (FASTA) and whether to show unique results only. The 'View' section below it shows the results are displayed in FASTA format. The main content area displays the FASTA sequences for several mitochondrial tRNA genes, including MT-TH, MT-TF, MT-TV, MT-TD, MT-TY, MT-TG, and MT-TW.

Dataset
Human genes (GRCh38.p13)

Filters
Gene type: Mt_tRNA

Attributes
Gene stable ID
Gene name
Unspliced (Transcript)

Dataset
[None Selected]

Export all results to ☐ Unique results only

Email notification to

View rows as ☐ Unique results only

>ENSG00000210176 | MT-TH
GTAAATATAGTTTAAACAAAACATCAGATTGTGAATCTGACAACAGAGGCTTACGACCCC
TTATTTACC
>ENSG00000210049 | MT-TF
GTTTATGTAGCTTACCTCCTCAAAGCAATACACTGAAAATGTTTAGACGGGCTCACATCA
CCCCATAAACA
>ENSG00000210077 | MT-TV
CAGAGTGTAGCTTAACACAAAGCACCCAACTTACACTTAGGAGATTTCAACTTAACTTGA
CCGCTCTGA
>ENSG00000210154 | MT-TD
AAGGTATTAGAAAAACCATTTTCTAACTTTGTCAAAGTTAAATTATAGGCTAAATCCTAT
ATATCTTA
>ENSG00000210144 | MT-TY
GGTAAATGGCTGAGTGAAGCATTGGACTGTAAATCTAAAGACAGGGGTTAGGCCTCTTT
TTACCA
>ENSG00000210164 | MT-TG
ACTCTTTTAGTATAAATAGTACCGTTAACTTCCAATTAAGTTTGTGACAACATTCAAA
AAAGAGTA
>ENSG00000210117 | MT-TW
AGAAATTTAGGTTAAATACAGACCAAGAGCCTTCAAAGCCCTCAGTAAGTTGCAATACTT
AATTTCTG

Use ENSEMBL BioMart – At “Attributes” choose “Sequences” and select the “Unspliced transcript”. At “Header information” select the Gene ID and the Gene Name and then on “Filters” select “Mt_tRNA” as the “Gene type”. Download the file (it will be in FASTA format) and put it in your working directory.

Where can I find the genomic coordinates of the nuclear tRNAs?

[Home](#) [Genomes](#) [Genome Browser](#) [Tools](#) [Mirrors](#) [Downloads](#) [My Data](#) [Projects](#) [Help](#) [About Us](#)

Table Browser

Use this program to retrieve the data associated with a track in text format, to calculate intersections between tracks, and to retrieve DNA sequence covered by a track. For help in using this application see [Using the Table Browser](#) for a description of the controls in this form, and the [User's Guide](#) for general information and sample queries. For more complex queries, you may want to use [Galaxy](#) or our [public MySQL server](#). To examine the biological function of your set through annotation enrichments, send the data to [GREAT](#). Refer to the [Credits](#) page for the list of contributors and usage restrictions associated with these data. All tables can be downloaded in their entirety from the [Sequence and Annotation Downloads](#) page.

clade: Mammal **genome:** Human **assembly:** Dec. 2013 (GRCh38/hg38)

group: Genes and Gene Predictions **track:** tRNA Genes [add custom tracks](#) [track hubs](#)

table: tRNAs [describe table schema](#)

region: ☒ genome ☐ position chr11:5,225,379-5,227,259 [lookup](#) [define regions](#)

identifiers (names/accessions): [paste list](#) [upload list](#)

filter: [create](#)

intersection: [create](#)

correlation: [create](#)

output format: BED - browser extensible data ☐ [Galaxy](#) ☐ [GREAT](#)

output file: (leave blank to keep output in browser)

file type returned: ☒ plain text ☐ gzip compressed

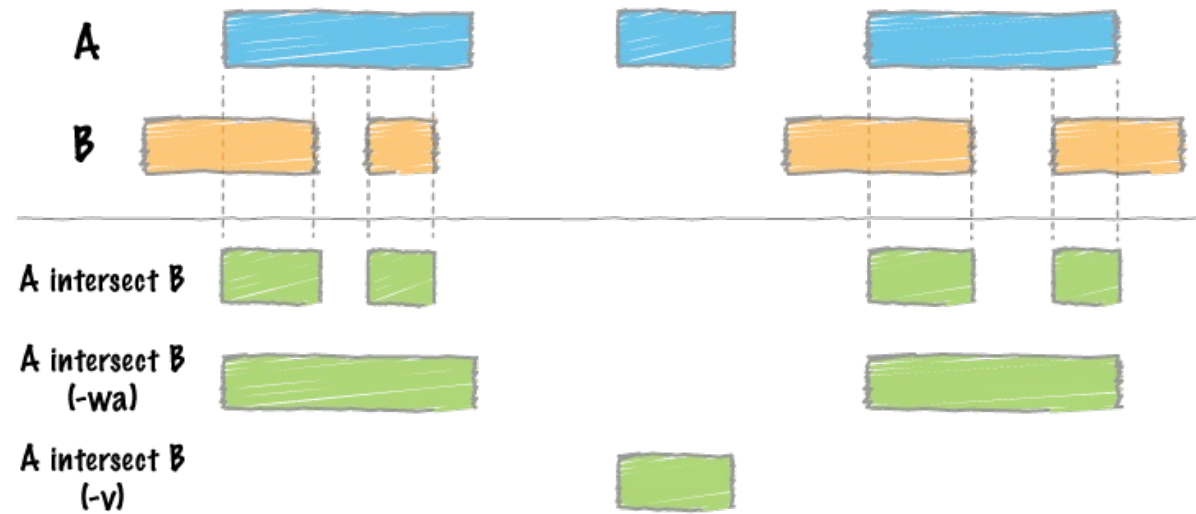
[get output](#) [summary/statistics](#)

To reset **all** user cart settings (including custom tracks), [click here](#).

Use the “Table Browser” tool of the UCSC Genome Browser (under the “Tools” menu). On the Human GRCh38 genome, select group “Genes and Predictions” and track “tRNA Genes”. Select the output format as “BED” and name the output file appropriately. On the next screen, make no changes and click “get BED”.

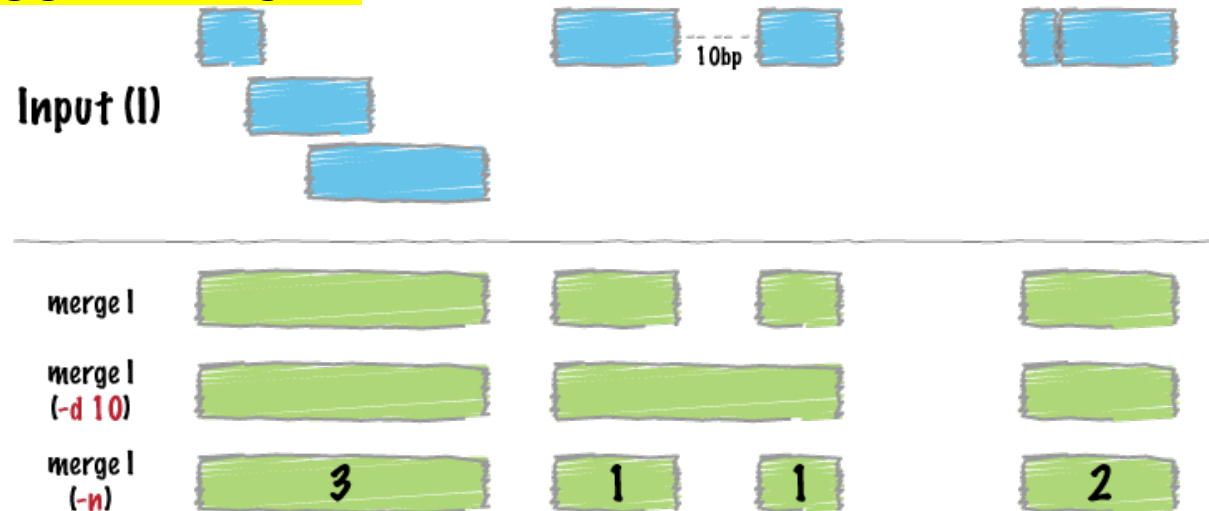
Note how this file does *not* contain the coordinates of the mitochondrial tRNAs. These tRNAs were computed with the tRNAscan-SE algorithm (for details click on the “describe table schema”) that computes genomic (i.e. nuclear) tRNAs and not mitochondrial ones.

Intersecting genomic regions



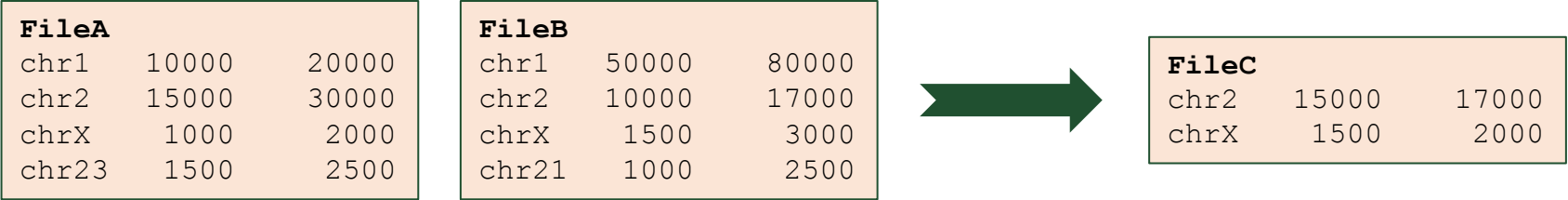
<https://bedtools.readthedocs.io/en/latest/content/tools/intersect.html>

Merging genomic regions

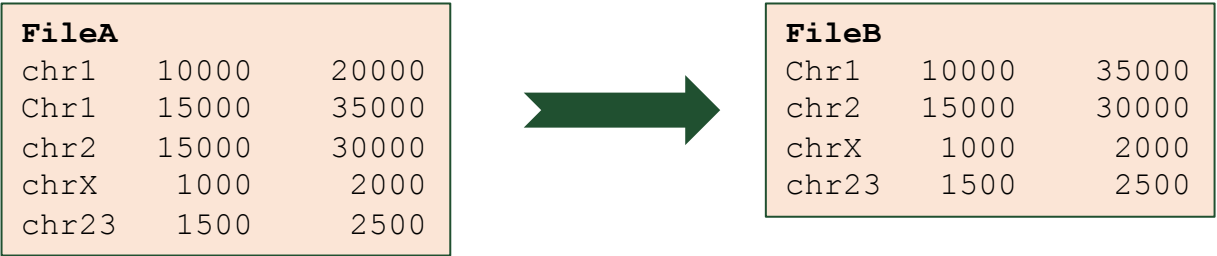


<https://bedtools.readthedocs.io/en/latest/content/tools/merge.html>

Intersecting genomic regions



Merging genomic regions



You should be ready to
start with **exercise 2**



Global vs. Local alignment

Global alignment

Query: GGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTG
||||| ||||| ||||| |||||
Search: GGTGGTCTAC--TTGGACCCAGAAGTTCTTTGAGGACTTTGGGGATCTG

Local alignment

Query: GGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTG
||| ||| ||| |||
Search: CACC--GAGATTCT-TGAAAT

For math/bioinformatic lovers:

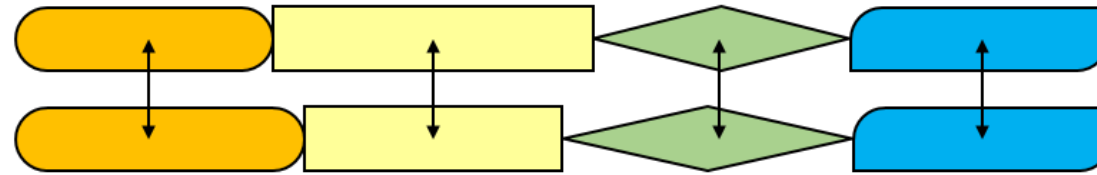
Needleman-Wunsch algorithm (for global alignments)
Smith-Waterman (for local alignments)

Needleman-Wunsch

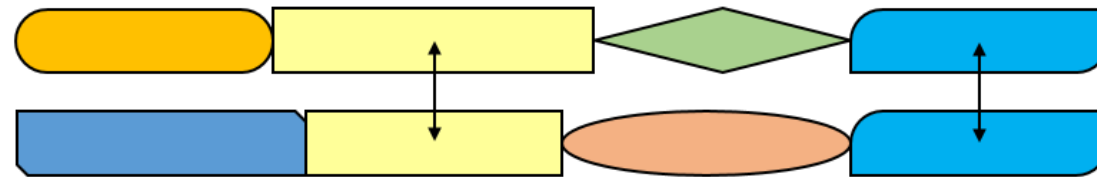
match = 1 mismatch = -1 gap = -1

		G	C	A	T	G	C	G
	0	-1	-2	-3	-4	-5	-6	-7
G	-1	1	0	-1	-2	-3	-4	-5
A	-2	0	0	1	0	-1	-2	-3
T	-3	-1	-1	0	2	1	0	-1
T	-4	-2	-2	-1	1	1	0	-1
A	-5	-3	-3	-1	0	0	0	-1
C	-6	-4	-2	-2	-1	-1	1	0
A	-7	-5	-3	-1	-2	-2	0	0

Global vs. Local alignment



Global Alignment



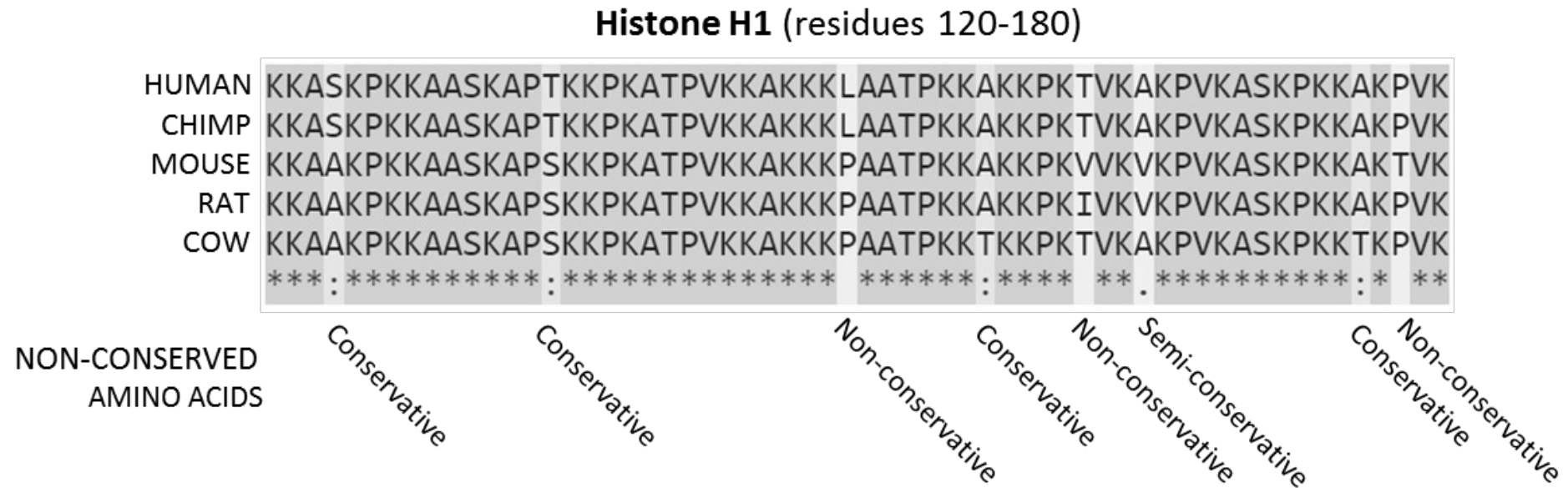
Local Alignment

By Yz cs5160 - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=54415549>

Global alignment: global optimization forcing alignment to span the entire length of the sequence(s)

Local alignment: identify similar smaller regions within longer sequences

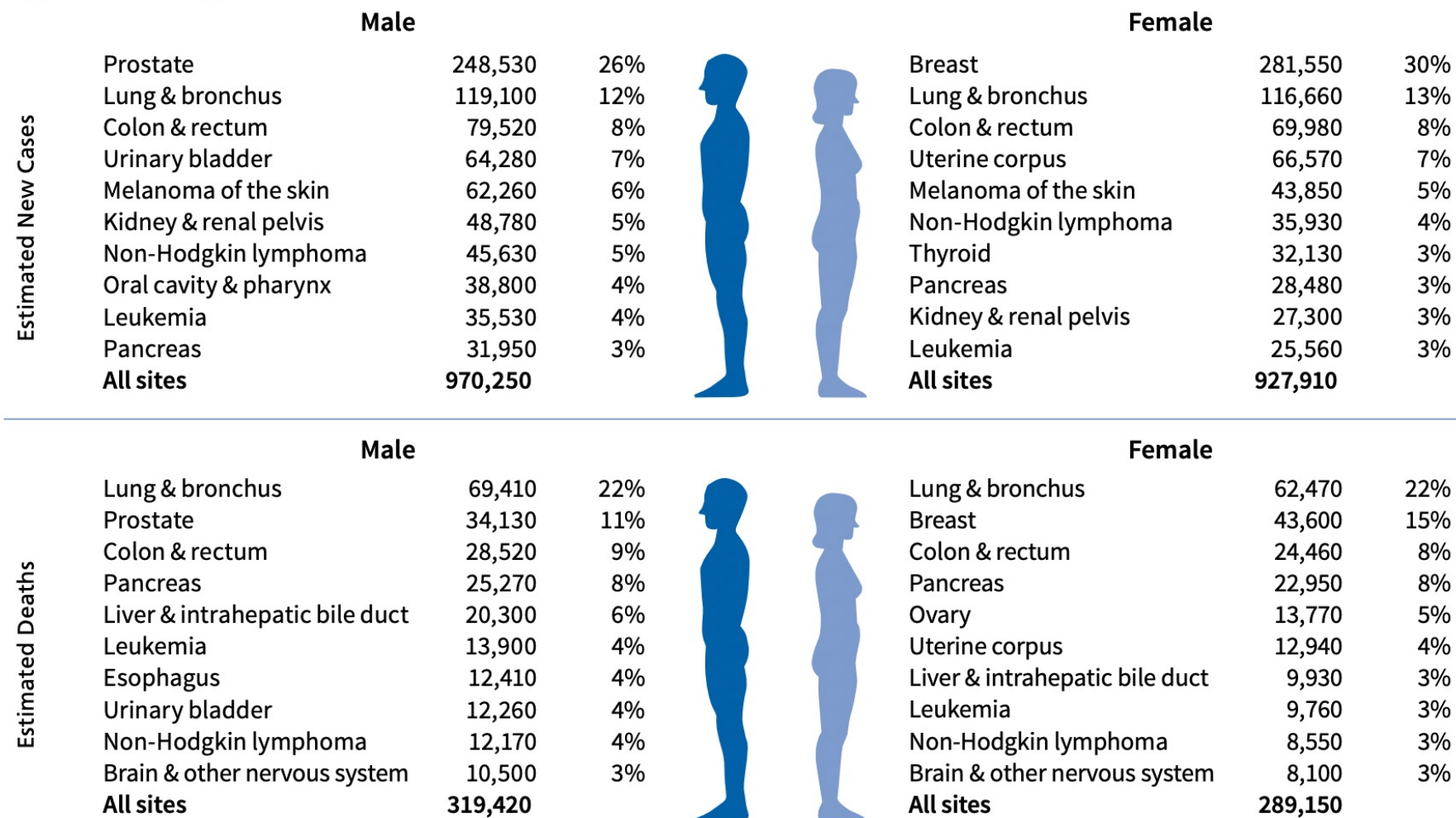
An example of global alignment



By Thomas Shafee - Own work, CC BY 4.0, <https://commons.wikimedia.org/w/index.php?curid=37188728>

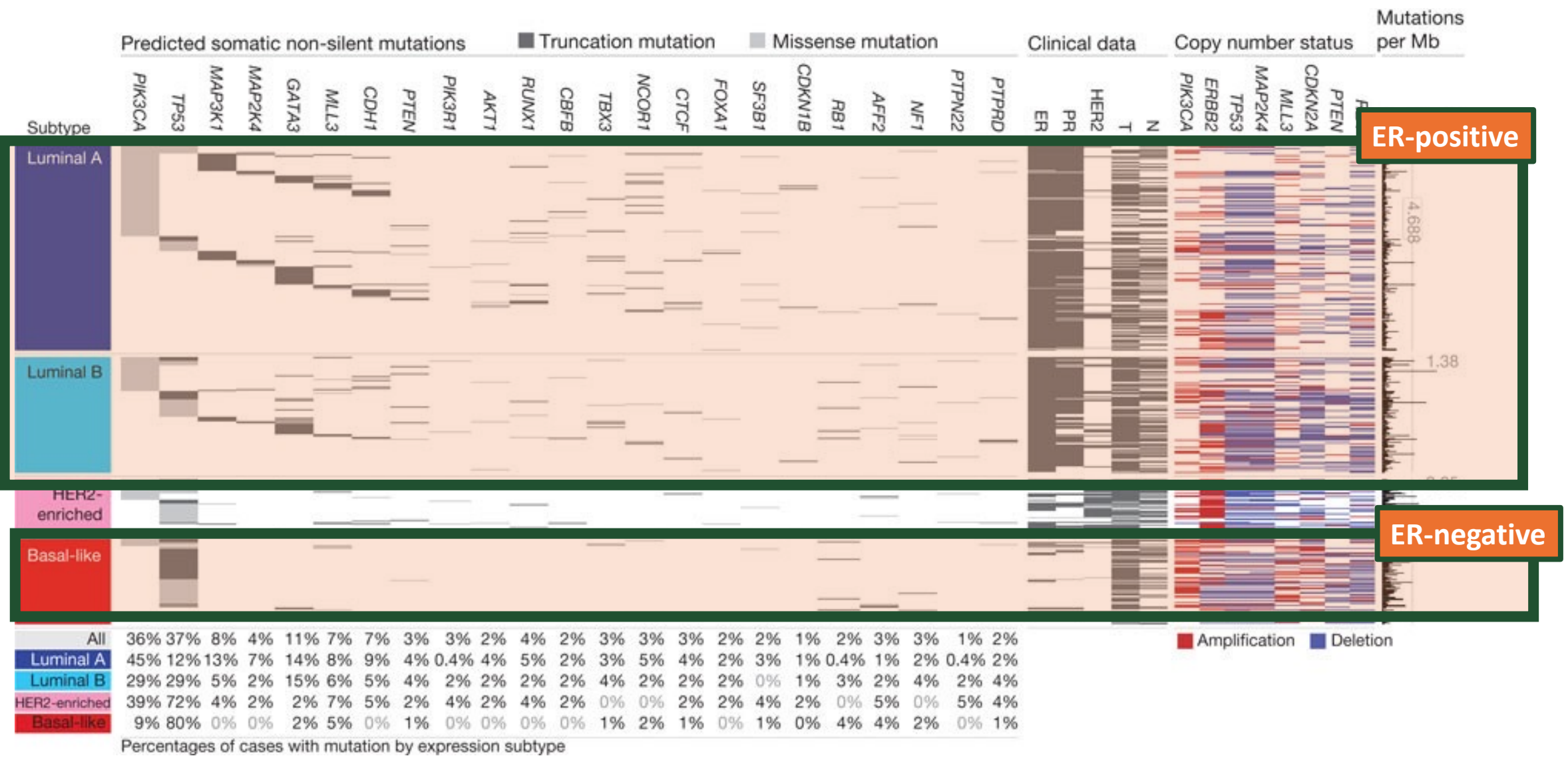
Breast cancer (BRCA): a deadly disease

Figure 3. Leading Sites of New Cancer Cases and Deaths – 2021 Estimates



Estimates are rounded to the nearest 10, and cases exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder. Estimates do not include Puerto Rico or other US territories. Ranking is based on modeled projections and may differ from the most recent observed data.

Analyzing RNA-seq data: Mapping on genome + Transcript quantification



ER-positive

ER-negative

The Cancer Genome Atlas Network (2012) Nature 490(7418):61-70

-- Our goal --

Compare the transcript profile of an ER-positive BRCA sample with an ER-negative BRCA sample

We need:

- ☐ A file with the sequences of nuclear chromosomes
- ☐ A file with the locations of genes and transcripts
- ☐ A tool to do the alignments
- ☐ A tool to do the transcript quantifications

Sources/Tools:

UCSC Genome browser

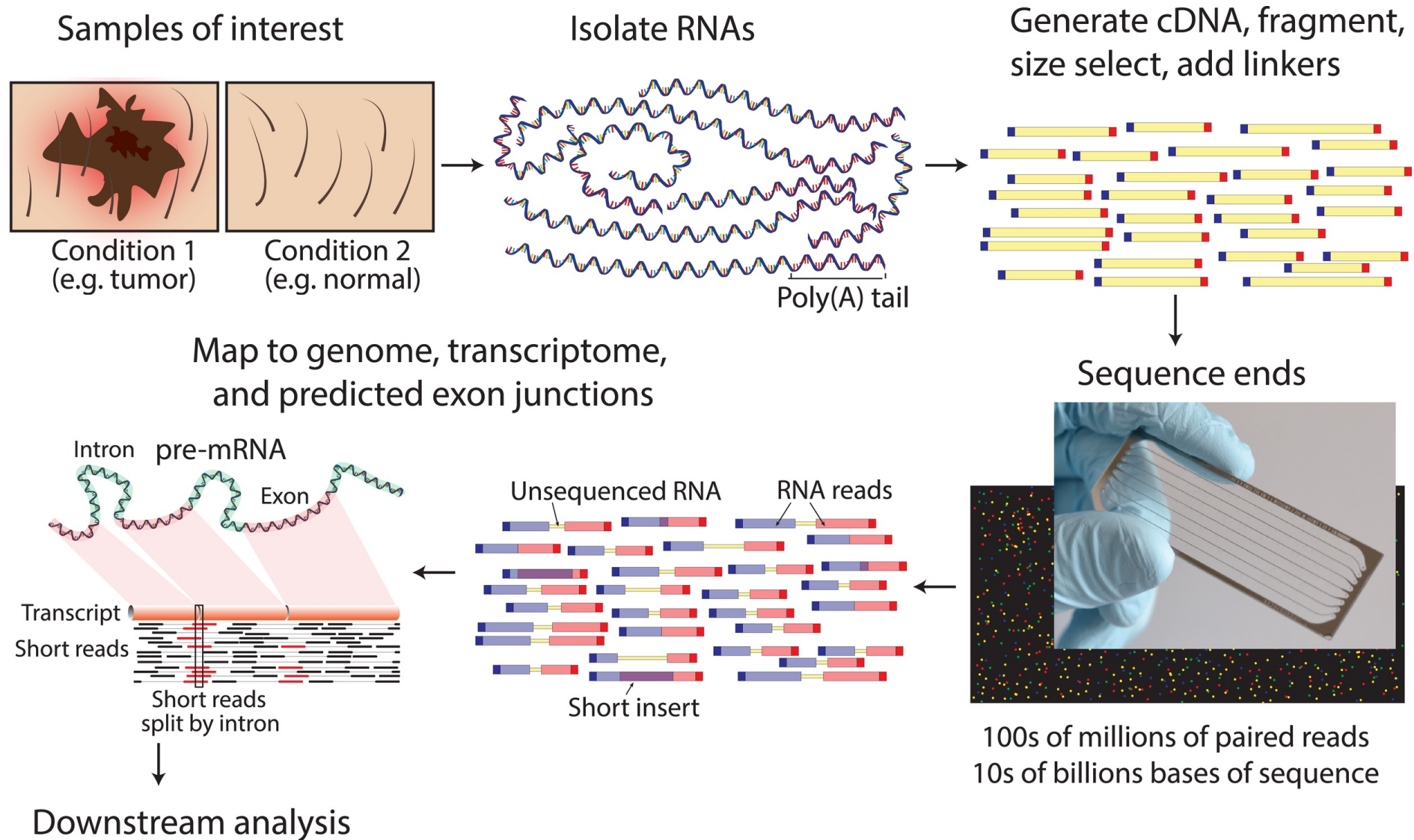
ENSEMBL/BioMart

STAR

RSEM

Note: due to large file sizes (>2 GB) and need for computational power, we will be working with chromosome 6

RNA sequencing experiments require aligning “reads” on the genome



By Malachi Griffith, Jason R. Walker, Nicholas C. Spies, Benjamin J. Ainscough, Obi L. Griffith -
<http://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1004393>, CC BY 2.5,
<https://commons.wikimedia.org/w/index.php?curid=53055894>

The FASTQ format

```
@A00773:150:HCC75DRXY:1:2101:1344:1031 1:N:0:TGCGGCGT+TACCGAGG
GGGGAAGCGCTACCGTAGATGCTCTTTCCTCCTGTGCCATCTCCCTGGTGAAGTCTCCGCCCTGGATCATGAAGTCCTTGATTACACGATGGAATTTGC
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00773:150:HCC75DRXY:1:2101:1452:1031 1:N:0:TGCGGCGT+TACCGAGG
CCTCGCTACTCCAATATGGCTTTAACCACCTCTTGGTGTCTCAGCTAAGAATGCCTGCCTCAGTTCAGCCTGGAAATCCACCACAGGTAAGTCTTGCTGCTGCG
+
F:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:,
@A00773:150:HCC75DRXY:1:2101:2663:1031 1:N:0:TGCGGCGT+TACCGAGG
CCCAGACCGATGTTCTTGTAGTACCGCGGGAGCTTCTCCTTGCCAGTTTCTCCCAGCAGGACCCTCTTCTTGTGTTTGAAGATGGTCGGCTGCTTTTGGTA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFF
@A00773:150:HCC75DRXY:1:2101:2736:1031 1:N:0:TGCGGCGT+TACCGAGG
CTCAGAGACACAAATATGAAGAAAGCACAGTCCCCAAAGCTTACTGATAGGCACAGGCTCTACACATCTGACAGAATGGGATGTGTATCTGAGACTAGTGA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00773:150:HCC75DRXY:1:2101:3007:1031 1:N:0:TGCGGCGT+TACCGAGG
GCGATGACCACGTTCCCATCTGGCTGGGTGCTGCAGAGGCTCAGCGGGAAGACCTTGGGGCTGGTCGGGGATGATGAGGAGACGGTGACCAGGGCTCCCTG
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00773:150:HCC75DRXY:1:2101:3115:1031 1:N:0:TGCGGCGT+TACCGAGG
CTGCTCCGTTTCCGACCTGGGCCGGTTCACCCCTCCTTAGGCAACCTGGTGGTCCCCCGCTCCCGGGAGGTCACCATATTGATGCCGAACCTTAGTGCGGAC
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00773:150:HCC75DRXY:1:2101:3260:1031 1:N:0:TGCGGCGT+TACCGAGG
ACTGGTAGTAGTAAAAGGGAAAACAGATAATAAGTTTTTAAAAAGTAAAATACAGAACATGTGAGTTATGGCATGTTAGAGGATGACAAGTAAAAGAAAAT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF:FFFFFFF,
@A00773:150:HCC75DRXY:1:2101:3405:1031 1:N:0:TGCGGCGT+TACCGAGG
CTAGTATTTCTTTGTCTTGTGTTTTTCCTTCAATTTAGAAGTGAGCATTGTGTTCTCAGCTATCAGAACTTTAAGCTGCCCACTATATTGAGATGCCCTTTTA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF::FFFFFFFFFFFFFFFFFFFFFFFF:FF
```

https://en.wikipedia.org/wiki/FASTQ_format

How to map billions of short reads onto genomes

Cole Trapnell & Steven L Salzberg

Mapping the vast quantities of short sequence fragments produced by next-generation sequencing platforms is a challenge. What programs are available and how do they work?

A new generation of DNA sequencers that can rapidly and inexpensively sequence billions of bases is transforming genomic science. These new machines are quickly becoming the technology of choice for whole-genome sequencing and for a variety of sequencing-based assays, including gene expression, DNA-protein interaction, human resequencing and RNA splicing studies¹⁻³. For example, the RNA-Seq protocol, in which processed mRNA is converted to cDNA and sequenced, is enabling the identification of previously unknown genes and alternative splice variants; the ChIP-Seq approach

Table 1 A selection of short-read analysis software

Program	Website	Open source?	Handles ABI color space?	Maximum read length
Bowtie	http://bowtie.cbcb.umd.edu	Yes	No	None
BWA	http://maq.sourceforge.net/bwa-man.shtml	Yes	Yes	None
Maq	http://maq.sourceforge.net	Yes	Yes	127
Mosaik	http://bioinformatics.bc.edu/marthlab/Mosaik	No	Yes	None
Novoalign	http://www.novocraft.com	No	No	None
SOAP2	http://soap.genomics.org.cn	No	No	60
ZOOM	http://www.bioinfor.com	No	Yes	240

Mapping sequencing data on the genome requires very good aligners

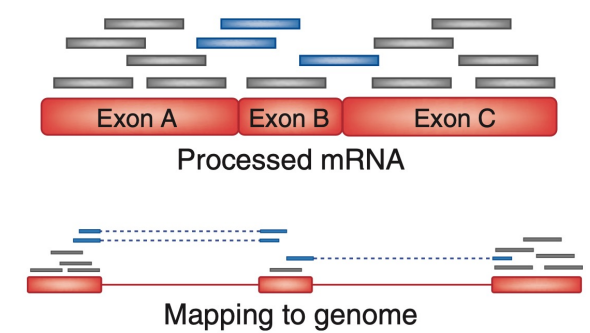
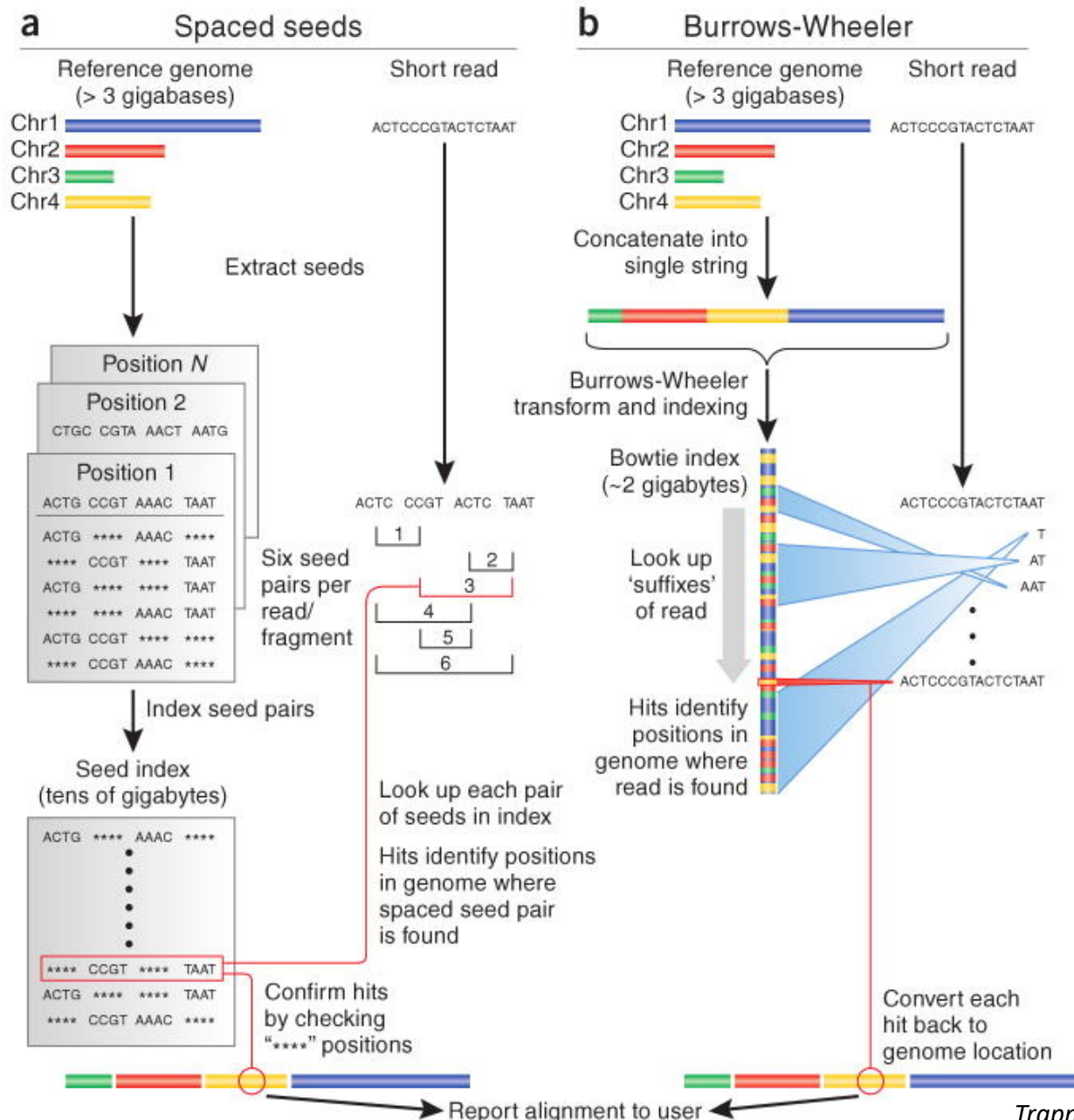


Figure 2 RNA-Seq assays produce short reads sequenced from processed mRNAs. Aligning these reads to the genome with Bowtie or Maq will produce the alignments shown in black but will fail to align the blue reads. A spliced-read mapper such as TopHat or ERANGE will also report the (blue) alignments spanning intron boundaries.

REVIEWS

APPLICATIONS OF NEXT-GENERATION SEQUENCING

Repetitive DNA and next-generation sequencing: computational challenges and solutions

Todd J. Treangen¹ and Steven L. Salzberg^{1,2}

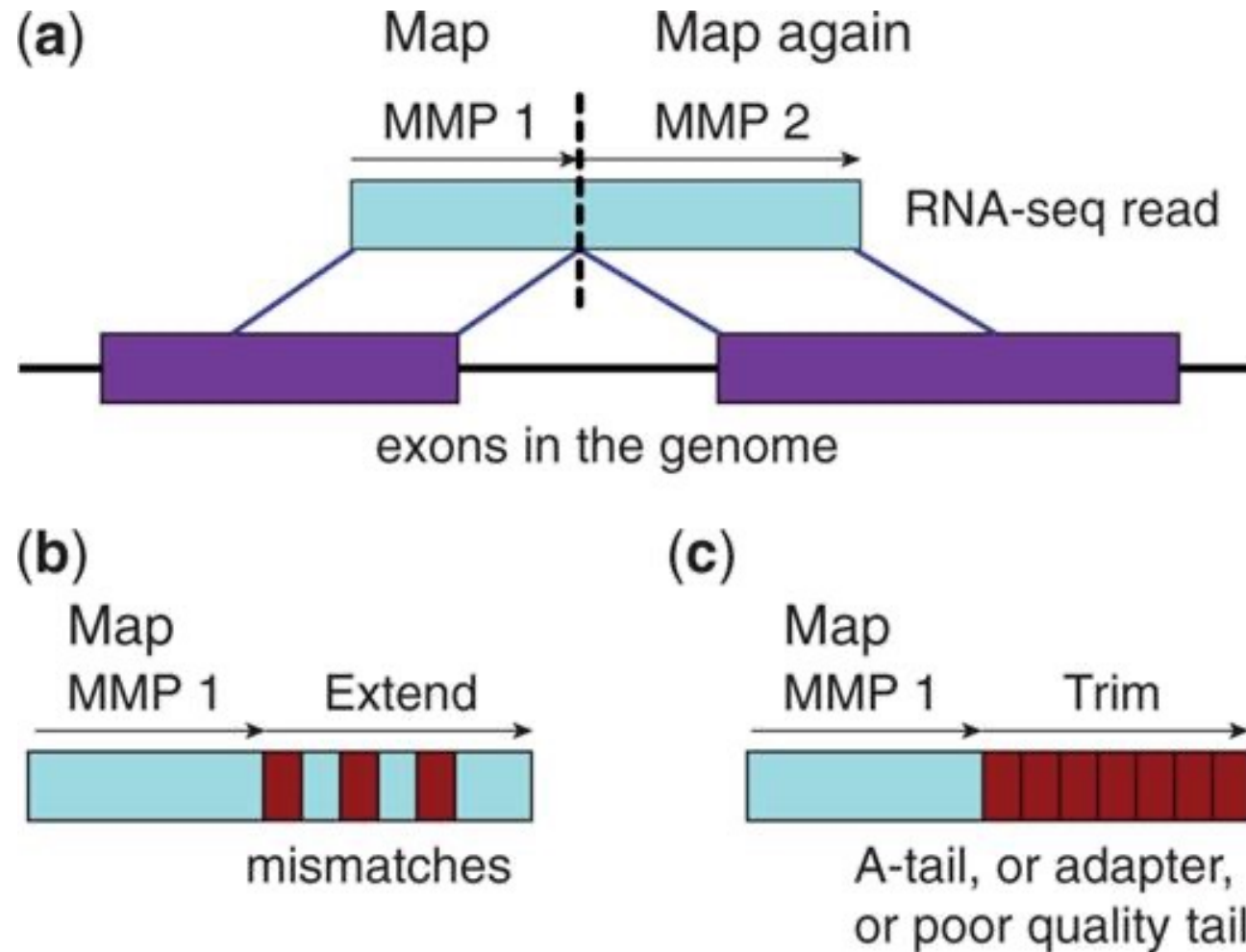
Abstract | Repetitive DNA sequences are abundant in a broad range of species, from bacteria to mammals, and they cover nearly half of the human genome. Repeats have always presented technical challenges for sequence alignment and assembly programs. Next-generation sequencing projects, with their short read lengths and high data volumes, have made these challenges more difficult. From a computational perspective, repeats create ambiguities in alignment and assembly, which, in turn, can produce biases and errors when interpreting results. Simply ignoring repeats is not an option, as this creates problems of its own and may mean that important biological phenomena are missed. We discuss the computational problems surrounding repeats and describe strategies used by current bioinformatics systems to solve them.

Mapping sequencing data on the genome requires very good aligners

Table 2 Overview of current computational tools for next-generation sequencing transcriptome analysis				
Scope	Program	Repeat-relevant parameters	Website	Refs
Spliced read alignment	GSNAP		http://share.gene.com/gmap	
	MapSplice		http://www.netlab.uky.edu/p/bioinfo/MapSplice	
	RUM		http://www.cbil.upenn.edu/RUM	
	SpliceMap		http://www.stanford.edu/group/wonglab/SpliceMap	
	TopHat		http://tophat.cbcb.umd.edu	
Reference-guided transcript assembly	Cufflinks	Improve repeat read mapping estimate: --multi-read-correct	http://cufflinks.cbcb.umd.edu	18,19
	ERANGE	Use multi-read fractions: --withmultifraction	http://woldlab.caltech.edu/rnaseq	5
	G-Mo.R-Se	None required; multi-reads supported by read aligner parameters	http://www.genoscope.cns.fr/externe/gmorse	
	Myrna	None required; multi-reads supported by read aligner parameters	http://bowtie-bio.sourceforge.net/myrna	46
	Scripture	None required; multi-reads supported by read aligner parameters	http://www.broadinstitute.org/software/scripture	
<i>De novo</i> transcript assembly	Multiple-k	None required or none available	http://www.surget-groba.ch/downloads	
	Rnnotator	None required or none available	None	
	Trinity	Separate transcripts derived from paralogues: --run_butterfly	http://trinityrnaseq.sourceforge.net	21
	Trans-ABYSS	None required or none available through command line	http://www.bcgsc.ca/platform/bioinfo/software/trans-abyss	76
	Velvet-Oases	Use long reads to resolve repeats: -long, -exp_cov auto	http://www.ebi.ac.uk/~zerbino/oases	77

The 'Program' column contains the name of program or algorithm. The 'Repeat-relevant parameters' column is a list of parameters that adjust how repeats are treated. The programs have many other parameters, but more careful treatment of repeats would start with modification of these.

STAR: probably the most popular RNA-seq aligner



Schematic representation of the Maximum Mappable Prefix search in the STAR algorithm for detecting (a) splice junctions, (b) mismatches and (c) tails

Mapping algorithms (almost) always return a sam (or bam) file

Format [\[edit \]](#)

The SAM format consists of a header and an alignment section.^[1] The binary equivalent of a SAM file is a [Binary Alignment Map](#) (BAM) file, which stores the same data in a compressed binary representation.^[4] SAM files can be analysed and edited with the software [SAMtools](#).^[1] The header section, if present, must precede the alignment section. Headings begin with the '@' symbol, which distinguishes them from the alignment section. Alignment sections have 11 mandatory fields, as well as a variable number of optional fields.^[1]

Col	Field	Type	Brief description
1	QNAME	String	Query template NAME
2	FLAG	Int	bitwise FLAG
3	RNAME	String	References sequence NAME
4	POS	Int	1- based leftmost mapping POSition
5	MAPQ	Int	MAPping Quality
6	CIGAR	String	CIGAR string
7	RNEXT	String	Ref. name of the mate/next read
8	PNEXT	Int	Position of the mate/next read
9	TLEN	Int	observed Template LENgth
10	SEQ	String	segment SEQuence
11	QUAL	String	ASCII of Phred-scaled base QUALity+33

[https://en.wikipedia.org/wiki/SAM_\(file_format\)](https://en.wikipedia.org/wiki/SAM_(file_format))

Analyzing RNA-seq data: Mapping on genome + Transcript quantification

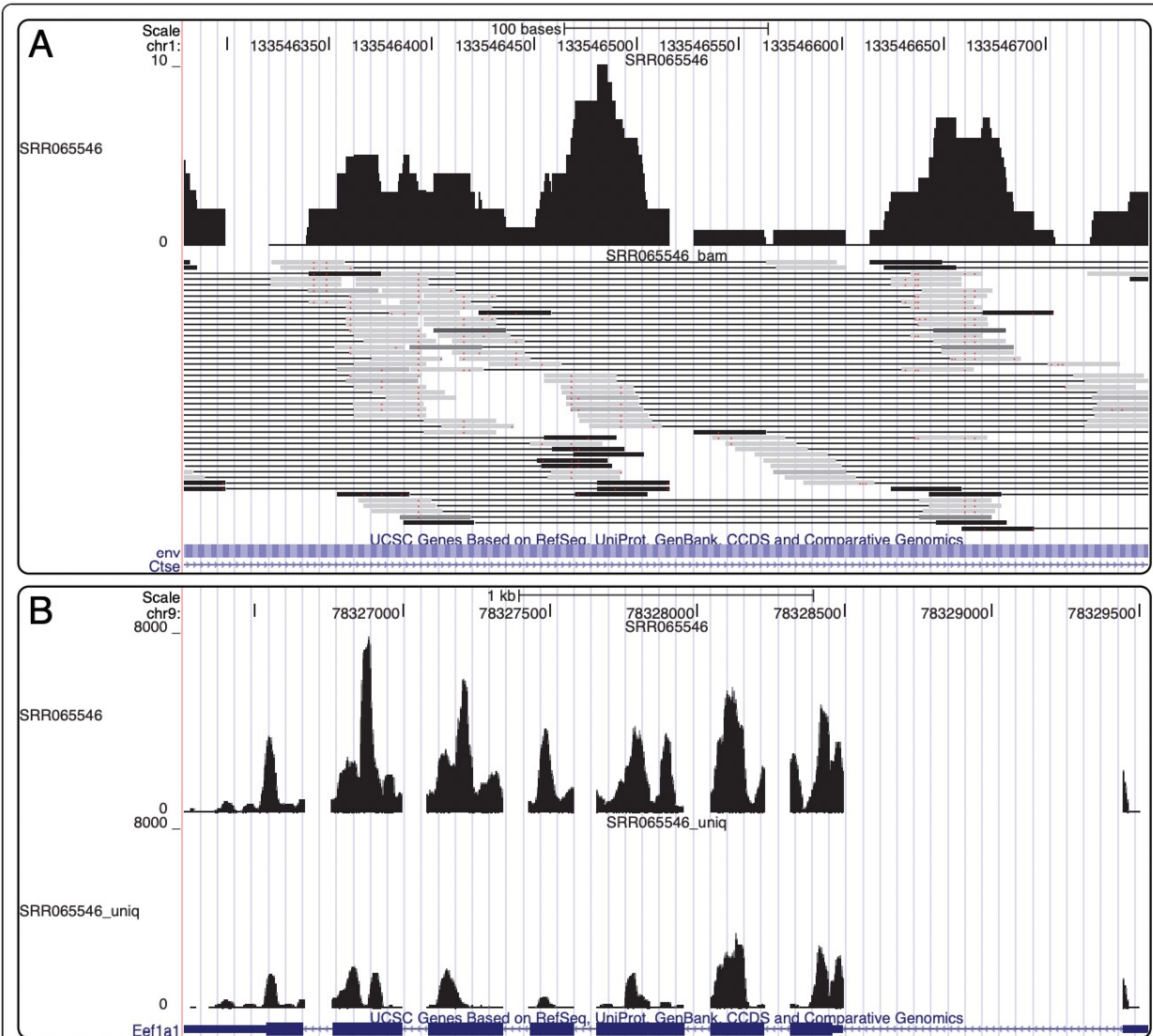


Figure 2 RSEM visualizations in the UCSC Genome Browser. Example visualizations of RSEM output from mouse RNA-Seq data set SRR065546 in the UCSC Genome Browser. (A) Simultaneous visualization of the wiggle output, which gives the expected read depth at each position in the genome, and the BAM output, which gives probabilistically-weighted read alignments. In the BAM track, paired reads are connected by a thin black line and the darkness of the read indicates the posterior probability of its alignment (black meaning high probability). (B) An example gene for which the expected read depth (top track) differs greatly from the read depth computed from uniquely-mapping reads only (bottom track).

Li and Dewey *BMC Bioinformatics* 2011, **12**:323
<http://www.biomedcentral.com/1471-2105/12/323>



SOFTWARE

Open Access

RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome

Bo Li¹ and Colin N Dewey^{1,2*}

Abstract

Background: RNA-Seq is revolutionizing the way transcript abundances are measured. A key challenge in transcript quantification from RNA-Seq data is the handling of reads that map to multiple genes or isoforms. This issue is particularly important for quantification with de novo transcriptome assemblies in the absence of sequenced genomes, as it is difficult to determine which transcripts are isoforms of the same gene. A second significant issue is the design of RNA-Seq experiments, in terms of the number of reads, read length, and whether reads come from one or both ends of cDNA fragments.

Results: We present RSEM, an user-friendly software package for quantifying gene and isoform abundances from single-end or paired-end RNA-Seq data. RSEM outputs abundance estimates, 95% credibility intervals, and visualization files and can also simulate RNA-Seq data. In contrast to other existing tools, the software does not require a reference genome. Thus, in combination with a de novo transcriptome assembler, RSEM enables accurate transcript quantification for species without sequenced genomes. On simulated and real data sets, RSEM has superior or comparable performance to quantification methods that rely on a reference genome. Taking advantage of RSEM's ability to effectively use ambiguously-mapping reads, we show that accurate gene-level abundance estimates are best obtained with large numbers of short single-end reads. On the other hand, estimates of the relative frequencies of isoforms within single genes may be improved through the use of paired-end reads, depending on the number of possible splice forms for each gene.

Conclusions: RSEM is an accurate and user-friendly software tool for quantifying transcript abundances from RNA-Seq data. As it does not rely on the existence of a reference genome, it is particularly useful for quantification with de novo transcriptome assemblies. In addition, RSEM has enabled valuable guidance for cost-efficient design of quantification experiments with RNA-Seq, which is currently relatively expensive.

You should be ready to
start with **exercise 3**

